Artur Iasenovets

Information Security Research Engineer

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Profile

4+ years of experience in software and security engineering. I develop in Go, Python, Rust, and TypeScript, with some experience in C++. I translate cryptographic protocols from papers into code, conduct security, correctness and leakage analysis, and develop systems for private data search, retrieval, and processing based on homomorphic encryption.

Education

Chongqing University of Posts and Telecommunications, Chongqing, China

- **M.S.** in Network and Information Security — Sep. 2024–Jul. 2027 *(expected)*
  - Supervisor: Prof. Tang Fei
  - GPA: 3.98/4.00

Ukhta State Technical University, Ukhta, Russia

- **B.S.** in Information Systems and Technologies — Mar. 2022–Jun. 2024
  - GPA: 4.49/5.00

St. Petersburg State University of Telecommunications, St. Petersburg, Russia

- **B.S. studies** in Information Systems and Technologies — Sep. 2020–Mar. 2022 *(transferred)*

Publications

Published Works and Preprints

Artur Iasenovets, Fei Tang, H. Zhu, P. Wang, and L. Liu. "Bringing Private Reads to Hyperledger Fabric via Private Information Retrieval," arXiv:2511.02656, 2025. Manuscript under review.

H. Zhu, F. Tang, W. Qin, J. Shan, P. Wang, and Artur Iasenovets. "EVMK-SSE: Efficient and Verifiable Multi-Keyword Symmetric Searchable Encryption," *Journal of King Saud University – Computer and Information Sciences*, 2025.

Artur Iasenovets and Fei Tang. "MFCTIIF: Multi-Feed Cyber Threat Intelligence Integration Framework," *International Research Journal*, no. 1 (163), 2026.

Piyumi M. S. Rajapaksha*, Artur Iasenovets*, Yinxue Yi, and Fei Tang. "TWAPPA-FRL: Trust-Weighted Privacy-Preserving Aggregation for Federated Reinforcement Learning in Collaborative Intrusion Detection," 2026. Manuscript under review. *Equal contribution.

Awards and Achievements

Chinese Government Scholarship (CSC) for Graduate Studies — 2024–2027
Full scholarship for the entire M.S. programme at CQUPT.

Sber Student Accelerator Regional Demo Day — 2023
Completed the Sber Student Accelerator with the AcademAI team and participated in the North-Western District regional demo day.

Winner, EdTech Project Competition — 2023
Presented LLM-based educational platform and won the competition organized by the MIPT Phystech School, the Fund for Development of Phystech Schools, and Startech.Base.

MIPT & Sber Phystech Gigachat Challenge — 2023
Participated in the Phystech Gigachat Challenge Hackathon, developing a LLM-based educational platform using GigaChat and Kandinsky APIs within a 48-hour sprint.

Kaspersky Cyberimmunity Hackathon 2.0 — 2023
Participated in Kaspersky's Cyberimmunity Hackathon in Fall 2023, placing 11th out of 100+ teams for developing traffic monitoring solutions for low-altitude airspace security.

First Degree Diploma, Severgeoecotech — 2022
Awarded first place in the Modern Information Technologies section of the 23rd International Youth Scientific Conference.

Supporting certificates

Experience

Graduate Researcher @ CQUPT — Sep. 2024–Present

- Designed and implemented BGV-based private information retrieval in Go using Lattigo for private world-state queries in Hyperledger Fabric, with query privacy under an HBC peer model based on the IND-CPA security of BGV.
- Designed the TWAPPA-FRL architecture and implemented CKKS-based private aggregation of DQN model updates using TenSEAL, with IND-CPA-based client-update privacy under an HBC aggregator model and explicit leakage analysis.
- Implemented and validated an FPS/RFPS-style private pattern-search prototype, replacing the DFT-based matching transform under approximate-arithmetic CKKS with an NTT-based matching transform under exact-arithmetic BGV/BFV, with IND-CPA-based pattern privacy under an HBC server model and explicit protocol leakage.
- Reconstructed preprocessing models, query construction, server computation, and response expansion of PIR protocols such as Piano, FrodoPIR/SimplePIR, XPIR, SealPIR, and OnionPIR, and analyzed their communication/performance trade-offs.

Information Security Engineer @ Positive Technologies — Aug. 2024–Feb. 2025

- Built an automated malware-processing pipeline with Airflow DAG that collects samples from MalwareBazaar, stores artifacts in S3, and scans them with YARA.
- Developed Python tools with unit-test coverage for validating Windows paths, network configurations, and endpoint-control policies.
- Analyzed EVTX logs, PowerShell, and malicious VBA macros; extracted IoCs and mapped TTPs to MITRE ATT&CK for CVE-2020-5902 and CVE-2023-38831.

Software Engineer @ AcademAI — Sep. 2023–Aug. 2024

- Developed the application layer with Next.js/TypeScript application, Python/Flask services, PostgreSQL/Prisma data layer, and Auth.js authentication for an LLM-based educational platform used by computer science university faculty and students.
- Integrated LangChain-based generation workflows with AsyncIO and Redis for asynchronous LLM orchestration and intermediate state management.
- Containerized the application with Docker, configured Nginx, deployed to cloud-hosted VPS infrastructure, and built the GitHub Actions CI/CD workflow.

Junior Software Engineer @ ZVEK “Progress” LLC — May. 2022–Aug. 2023

- Built a C++17/Crow REST backend with MySQL and Qt 5/QML desktop and mobile clients for a QR-based warehouse inventory system.
- Supported approximately 1,000 records and 20+ concurrent users; added unit-test coverage, CMake builds, Git-based development, debugging, and Linux deployment.

Skills

- **Programming:** Go, Python, Rust, TypeScript, C++, SQL, Bash, LaTeX
- **Privacy-enhancing technologies:** Private Information Retrieval (PIR/CPIR), Searchable Symmetric Encryption (SSE), Private Pattern Search
- **Homomorphic encryption:** Brakerski–Gentry–Vaikuntanathan (BGV), Brakerski–Fan–Vercauteren (BFV), Cheon–Kim–Kim–Song (CKKS)
- **Post-quantum cryptography:** ML-KEM, ML-DSA, SLH-DSA; LWE, RLWE
- **Cryptographic libraries:** Lattigo, SEAL, TenSEAL, OpenFHE, fheanor
- **Systems and infrastructure:** Linux, Proxmox VE, VMware, Docker, GitHub Actions, GitLab CI/CD, Nginx, Ansible, Airflow, S3, Hyperledger Fabric
- **Security engineering:** YARA, Chainsaw, malware/log analysis, MITRE ATT&CK
- **Languages:** Russian — native; English — C1; Chinese — HSK3